

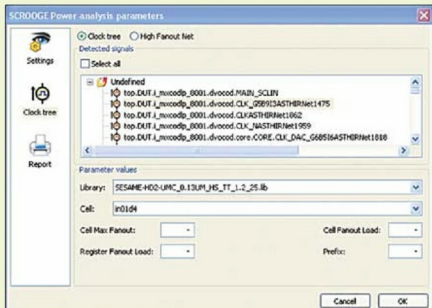


Fig. 3: Easy setup available in the SCROOGE tool

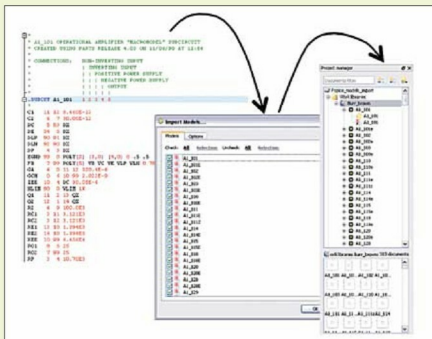


Fig. 4: Import SPICE model libraries and autocreate symbols for direct use with SLED tool

combining SPICE and HDL representations to ease the architectural design of any system-on-chip (SoC). A system can be iteratively validated by using a SPICE representation of a single block, while all other blocks in the system are represented in HDL for a fully-functional mixed-signal simulation. It is fundamental for state-of-the-art hierarchical designs of multi-domain components. For design of analogue

circuits, the generic library of SPICE components combined with multi-flavour SPICE netlisting improves designers' productivity by enabling creation of true mixed-signal circuits.

Ease of use. SLED has a direct syntax coloured behavioural code display to facilitate reading of HDL and HDL-AMS models. It comes with easy push/pop navigation through a list of hierarchical schematics. It has click-'n'-drop

symbols of any blocks from any generic or ad-hoc library and drags them while preserving wire connections.

Schematic consistency checks. SLED saves time due to early error detection during design with coherency checks, i.e., symbol vs model I/O, and online electrical rule checks, i.e., shorted output pins.

Interoperability and compatibility. The design data—including design or reference library configuration files, symbols, schematics—is stored in an ASCII format, enabling a script-based approach for automatic symbol generation, library conversion and design library setup. It comes with a built-in handling of design context configurations for multi-flavour SPICE netlisting (SMASH, Hspice, SDL, LVS).

EDIF import/export. The option to import/export to and from EDIF provides the means to exchange full schematics including graphics with other design entry solutions, as well as to transfer designs to the next step in your design chain, such as schematic-driven layout.

Design of multi-domain systems. SLED offers the capability to create highly flexible and complex models through graphic composition. It covers the modelling of any domain like analogue electronics, digital electronics, mechanics, magnetic and thermal effects. SLED is ideal for multi-domain systems where designers are guided with intelligent-type checking of terminals during wiring of symbols. Initialising homogeneous simulation of such heterogeneous systems in SMASH avoids simulator couplings with its disadvantages in configuration, setup and convergence problems.

SCROOGE power consumption estimator

SCROOGE is a power consumption analyser for simulating mixed-signal power consumption hierarchically. Where the common solutions just enable statistical or average power consumption analysis, the USP of SCROOGE is in providing a hierarchical evaluation of power consumption.